Department of Computer Science & Engineering

CSCI/CSCY 4407: Security & Cryptography

Lab 5 Report: MAC, UF-CMA, CBC-MAC, HMAC & PRF

Group Number: Group 10 **Semester:** Spring 2026 **Instructor:** Dr. Victor Kebande **Teaching Assistant:** Celest Kester **Submission Date:** 03/20/2026

Group Members:

- Matthew Kenner
- Jonathan Le
- Cassius Kemp

Table of Contents

1. Task 1 – Lab Setup and Directory Creation
2. Task 2 – Create and Inspect Sample Message Files
3. Task 3 – Write and Test a Basic HMAC Generation Script
4. Task 4 – Verify the Authenticity of a Message
5. Task 5 – Modify the Message and Test Integrity Failure
6. Task 6 – Generate HMAC Tags for Multiple Messages
7. Task 7 – Change the Secret Key and Observe the Effect
8. Task 8 – Replay Attack Experiment
9. Task 9 – Implementing a PRF-Based Authentication Function

Introduction

This report documents the implementation and analysis performed for Lab 5 on message authentication codes (MACs), HMAC, replay attacks, and PRF-based authentication. Each task was completed in a Linux environment using Python 3 and standard cryptographic libraries. The report includes commands executed, source code, terminal outputs, screenshots, and interpretations of results for each experiment.

Environment

All experiments were performed in a Linux environment using Kali Linux. Python 3 was used for all scripts, and SHA-256/HMAC operations were implemented using Python's built-in `hashlib` and `hmac` libraries.

- **Operating System:** Kali Linux
- **Python Version:** Python 3.12
- **Terminal:** Kali Linux terminal
- **Installation:** Local Kali Linux install

Files Included

The following Python source files are included in this submission:

- `hmac_gen.py` — Task 3: HMAC generation for a single message
- `hmac_verify.py` — Task 4: HMAC verification script
- `hmac_multi.py` — Task 6: HMAC generation for multiple messages
- `prf_experiment.py` — Task 9: PRF-based authentication function

Task 1 – Lab Setup and Directory Creation

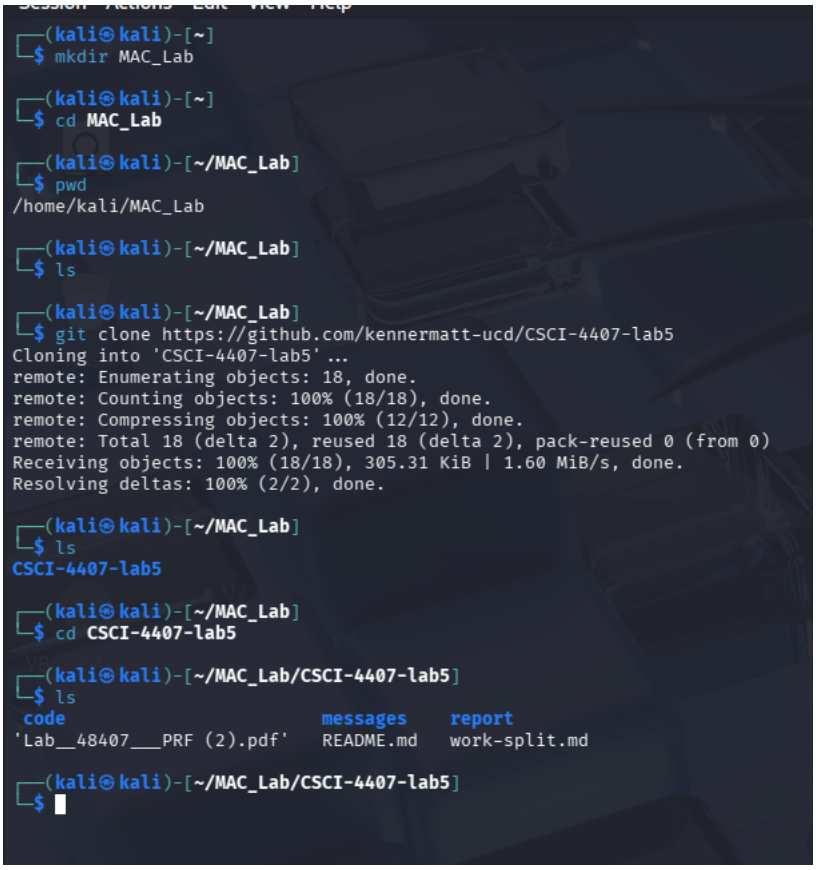
What Was Done

A dedicated working directory named `CSCI-4407-lab5` was created to organize all scripts, message files, and experimental outputs for this lab. The `mkdir` command created the folder, `cd` navigated into it, and `pwd` confirmed the current working path.

Commands Executed

```
mkdir MAC_Lab # created and pulled from github
cd MAC_Lab
git clone https://github.com/kennermatt-ucd/CSCI-4407-lab5.git
cd CSCI-4407-lab5
pwd
ls
```

Screenshot Evidence



> Shows terminal output of `mkdir CSCI-4407-lab5`, `cd MAC_Lab`, `pwd` displaying the full path, and `ls` confirming the directory is empty.

Explanation

The working directory serves as the root location for all lab files. Having a dedicated folder prevents accidental overwriting of unrelated files and makes it straightforward to reproduce the experiment from a clean state. The `pwd` command confirmed we were in the correct directory before proceeding to any subsequent tasks. The team also used a shared GitHub repository to coordinate scripts, screenshots, and report drafting, although GitHub setup was not a required graded component of Task 1.

Task 2 – Create and Inspect Sample Message Files

What Was Done

Three plaintext message files were created to represent realistic messages that a sender might transmit to a receiver. These files serve as the input to the authentication experiments in later tasks. The SHA-256 hash of each file was computed to demonstrate the avalanche effect: even a tiny change in a message produces a completely different hash value.

Commands Executed

```
echo "Transfer 100 dollars to Bob" > message1.txt
echo "Transfer 500 dollars to Alice" > message2.txt
echo "Authorize access to server room" > message3.txt

cat message1.txt
cat message2.txt
cat message3.txt

ls -l message*.txt

sha256sum message1.txt
sha256sum message2.txt
sha256sum message3.txt

echo "Transfer 1000 dollars to Bob" > message1.txt
sha256sum message1.txt
```

Screenshot Evidence

```
┌──(kali㉿kali)-[~/MAC_Lab/CSCI-4407-lab5]
└─$ echo "Transfer 100 dollars to Bob" > messages/message1.txt

┌──(kali㉿kali)-[~/MAC_Lab/CSCI-4407-lab5]
└─$ echo "Transfer 500 dollars to Alice" > messages/message2.txt

┌──(kali㉿kali)-[~/MAC_Lab/CSCI-4407-lab5]
└─$ echo "Authorize access to server room" > messages/message3.txt
```

```
┌──(kali㉿kali)-[~/MAC_Lab/CSCI-4407-lab5]
└─$ cd messages

┌──(kali㉿kali)-[~/MAC_Lab/CSCI-4407-lab5/messages]
└─$ cat message1.txt
Transfer 100 dollars to Bob

┌──(kali㉿kali)-[~/MAC_Lab/CSCI-4407-lab5/messages]
└─$ cat message2.txt
Transfer 500 dollars to Alice

┌──(kali㉿kali)-[~/MAC_Lab/CSCI-4407-lab5/messages]
└─$ cat message3.txt
Authorize access to server room
```

> Shows the three `echo` commands creating the message files and confirmation via `cat`.

```
┌──(kali㉿kali)-[~/MAC_Lab/CSCI-4407-lab5/messages]
└─$ ls -l message*.txt
-rw-rw-r-- 1 kali kali 28 Mar 19 18:13 message1.txt
-rw-rw-r-- 1 kali kali 30 Mar 19 18:13 message2.txt
-rw-rw-r-- 1 kali kali 32 Mar 19 18:14 message3.txt
```

> Shows `ls -l message*.txt` output with file sizes.

```
┌──(kali㉿kali)-[~/MAC_Lab/CSCI-4407-lab5/messages]
└─$ sha256sum message1.txt
cd6bafd58bffad3e22a31799be12860fb5add65c3fd2a9b9bfd562f035fd97c1  message1.txt

┌──(kali㉿kali)-[~/MAC_Lab/CSCI-4407-lab5/messages]
└─$ sha256sum message2.txt
8e6d7f11d2e3185eb205037d9bb5fde4b21cae575222cb5fd48f54de17167cd1  message2.txt

┌──(kali㉿kali)-[~/MAC_Lab/CSCI-4407-lab5/messages]
└─$ sha256sum message3.txt
a1a3d085142fb75c2e940b9e35e904cbef83c60ba14520576e775ff26d4d624f  message3.txt
```

> Shows the SHA-256 hash values for all three original messages.

```
┌──(kali㉿kali)-[~/MAC_Lab/CSCI-4407-lab5/messages]
└─$ echo "Transfer 1000 dollars to Bob" > message1.txt

┌──(kali㉿kali)-[~/MAC_Lab/CSCI-4407-lab5/messages]
└─$ sha256sum message1.txt
83474ade2dc54f59df69d10e50d6e090f72b8d1fc5bbbbcd87c403b262c7869d  message1.txt

┌──(kali㉿kali)-[~/MAC_Lab/CSCI-4407-lab5/messages]
└─$
```

> Shows the recomputed hash of `message1.txt` after the amount was changed from 100 to 1000.

Recorded Hash Values (Original)

File	SHA-256 Hash
message1.txt	cd6bafdb58bffad3e22a31799be12860fb5add65c3fd2a9b9bfd562f035fd97c1
message2.txt	8e6d7f11d2e3185eb205037d9bb5fde4b21cae575222cb5fd48f54de17167cd1
message3.txt	a1a3d085142fb75c2e940b9e35e904cbef83c60ba14520576e775ff26d4d624f

Recorded Hash After Modification

"Transfer 1000 dollars to Bob"

File	SHA-256 Hash (Modified)
message1.txt	83474ade2dc54f59df69d10e50d6e090f72b8d1fc5bbbbcd87c403b262c7869d

Explanation

The SHA-256 hash function is a one-way cryptographic function that maps any input to a fixed-length 256-bit output. When the amount in `message1.txt` was changed from "100" to "1000", the resulting hash was completely different from the original. This demonstrates the **avalanche effect**: even a very small change in the message produces a completely different hash output. This property is fundamental to detecting message tampering, because any modification to a message will invalidate its hash or authentication tag.

Task 3 – Write and Test a Basic HMAC Generation Script

What Was Done

A Python script named `hmac_gen.py` was created to compute an HMAC authentication tag for a message. HMAC uses a shared secret key combined with a cryptographic hash function (SHA-256) to produce a tag that can only be reproduced by someone who knows the key. The script was run three times: once for each

message file. Between runs, the `filename` variable was changed from `message1.txt` to `message2.txt` and then to `message3.txt` to generate tags for each file separately.

Script: hmac_gen.py

```
import hmac
import hashlib
import sys
import os

key = b"secretkey123"

script_dir = os.path.dirname(os.path.abspath(__file__))
messages_dir = os.path.join(script_dir, "..", "messages")

filename = input("Enter message filename (e.g. message1.txt): ").strip()
filepath = os.path.join(messages_dir, filename)

with open(filepath, "rb") as f:
    message = f.read()

tag = hmac.new(key, message, hashlib.sha256).hexdigest()

print("Message:", message.decode())
print("HMAC Tag:", tag)
```

Commands Executed

```
# Run 1 — filename = "message1.txt"
python3 hmac_gen.py

# Run 2 — filename changed to "message2.txt"
python3 hmac_gen.py

# Run 3 — filename changed to "message3.txt"
python3 hmac_gen.py

# Run 4 — message1.txt modified, filename = "message1.txt"
python3 hmac_gen.py
```

Screenshot Evidence

alt text

Recorded HMAC Tags

File	HMAC-SHA256 Tag (key: secretkey123)
message1.txt	223e2b14c5fe1d878c436101936543d5b81bc55a20c94bc30d6cf2ba284fee59
message2.txt	b94ec115fa380925e2a2272ac479b9ed83b9d689148961d2c870a636d639c752
message3.txt	cb0bd950268a87b8a73a514c951d20b99bcd9af9f962ea53f60ed09e0b7acbcc
message1.txt (modified)	98355bb8b5bbc50272e48a98a03537aeb5f2d4a50d7be5cb2055c3d5715b4f85

Step-by-Step Explanation

The script operates as follows:

1. The secret key `b"secretkey123"` is defined as a byte string. This key is shared between the sender and receiver.
2. The message file is opened and read in binary mode to ensure all byte values are handled correctly.
3. `hmac.new(key, message, hashlib.sha256)` constructs an HMAC object using SHA-256 as the underlying hash function, combining the key and message according to the HMAC construction: `HMAC(K, M) = H((K ⊕ opad) || H((K ⊕ ipad) || M))`.
4. `.hexdigest()` returns the authentication tag as a hexadecimal string, which is the value `T = T_K(M)`.
5. Both the message and tag are printed to the terminal.

Why Different Messages Produce Different HMAC Values

Even though the same secret key is used for all three messages, each message produces a completely different HMAC tag. This occurs because the HMAC function combines both the key and the message as inputs before hashing. Since the input to the hash function changes with every different message, the output also changes. This is a required property of a secure MAC: distinct messages must produce distinct tags, so that an attacker cannot substitute one authenticated message for another.

Effect of Modifying a Message

When `message1.txt` was modified slightly (for example, changing the dollar amount), the resulting HMAC tag changed entirely. This confirms the avalanche effect operates through the HMAC computation: even a one-character difference in the message causes the tag to be completely unrelated to the original. This behavior is what allows a receiver to detect any tampering with an authenticated message.

Task 4 – Verify the Authenticity of a Message

What Was Done

A Python verification script `hmac_verify.py` was implemented to check whether a received message-tag pair is valid. The script recomputes the HMAC tag from the current file contents using the same secret key and uses `hmac.compare_digest` for a timing-safe comparison against the provided tag. Four scenarios were tested: correct tag, wrong tag, modified message with original tag, and verification repeated for `message2.txt`.

Script: hmac_verify.py

```python
import hmac
import hashlib

key = b"secretkey123"
filename = "message1.txt"
provided_tag = input("Enter the received HMAC tag: ").strip()

with open(filename, "rb") as f:
    message = f.read()

computed_tag = hmac.new(key, message, hashlib.sha256).hexdigest()

if hmac.compare_digest(provided_tag, computed_tag):
    print("Verification successful: Message is authentic.")
else:
    print("Verification failed: Message has been altered or tag is invalid.")
```

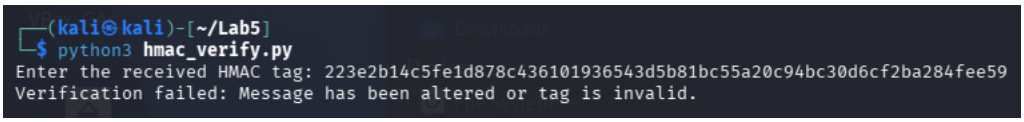
Commands Executed

```
python3 hmac_verify.py
# Test 1: entered the correct HMAC tag from Task 3 → expected: success
# Test 2: entered the tag with one character changed → expected: failure
# Test 3: modified message1.txt, then entered the original tag → expected: failure
# Test 4: changed filename to message2.txt, entered correct tag → expected: success
# Test 5: modified message2.txt, then entered the original tag → expected: failure
# Test 6: entered the tag with one character changed → expected: failure
```

Screenshot Evidence

> Shows the script accepting the correct HMAC tag and printing "Verification successful: Message is authentic."

> Shows the script rejecting an incorrect tag (one character changed) and printing the failure message.



> Shows the script rejecting the original tag when the message file was modified.

```
┌──(kali㉿kali)-[~/Lab5]
└─$ python3 hmac_verify.py
Enter the received HMAC tag: b94ec115fa380925e2a2272ac479b9ed83b9d689148961d2c870a636d639c752
Verification successful: Message is authentic.
```

```
┌──(kali㉿kali)-[~/Lab5]
└─$ python3 hmac_verify.py
Enter the received HMAC tag: b94ec115fa380925e2a2272ac479b9ed83b9d689148961d2c870a636d639c752
Verification failed: Message has been altered or tag is invalid.
```

```
┌──(kali㉿kali)-[~/Lab5]
└─$ python3 hmac_verify.py
Enter the received HMAC tag: b94ec115fa380925e2a2272ac479b9ed83b9d689148961d2c870a636d639c700
Verification failed: Message has been altered or tag is invalid.
```

> Shows verification repeated for `message2.txt` with its correct tag.

Explanation

HMAC ensures integrity as any modification to the message not matter the size (can even be a single bit) will give us a totally different tag. HMAC ensures authenticity as the tage is generated using a secret key known onlt to the two parties engaged in communication (sender and reciever), any attacker can not forge the message without the key needed to generate a valid tag.

Task 5 – Modify the Message and Test Integrity Failure

What Was Done

A message tampering experiment was conducted to demonstrate that a MAC detects unauthorized modifications. The original HMAC tag from Task 3 was used as the reference. The message file was modified three times, and the verification script was run after each modification using the original tag. All three attempts failed verification.

Commands Executed

```
# Modification 1
echo "Transfer 9000 dollars to Eve" > message1.txt
cat message1.txt
python3 hmac_verify.py

# Modification 2
echo "Transfer 100 dollars to Eve" > message1.txt
cat message1.txt
python3 hmac_verify.py

# Modification 3
echo "Transfer 100 dollars to Bob immediately" > message1.txt
cat message1.txt
python3 hmac_verify.py
```

Screenshot Evidence

```
┌──(kali㉿kali)-[~/Lab5]
└─$ echo "Transfer 9000 dollars to Eve" > message1.txt

┌──(kali㉿kali)-[~/Lab5]
└─$ cat message1.txt
Transfer 9000 dollars to Eve

┌──(kali㉿kali)-[~/Lab5]
└─$ python3 hmac_verify.py
Enter the received HMAC tag: 223e2b14c5fe1d878c436101936543d5b81bc55a20c94bc30d6cf2ba284fee59
Verification failed: Message has been altered or tag is invalid.
```

> Shows the first modified message content and the verification failure output.

```
┌──(kali㉿kali)-[~/Lab5]
└─$ echo "Transfer 100 dollars to Eve" > message1.txt

┌──(kali㉿kali)-[~/Lab5]
└─$ cat message1.txt
Transfer 100 dollars to Eve

┌──(kali㉿kali)-[~/Lab5]
└─$ python3 hmac_verify.py
Enter the received HMAC tag: 223e2b14c5fe1d878c436101936543d5b81bc55a20c94bc30d6cf2ba284fee59
Verification failed: Message has been altered or tag is invalid.
```

> Shows the second modification and verification failure.

```
┌──(kali㉿kali)-[~/Lab5]
└─$ echo "Transfer 100 dollars to Bob immediately" > message1.txt

┌──(kali㉿kali)-[~/Lab5]
└─$ cat message1.txt
Transfer 100 dollars to Bob immediately

┌──(kali㉿kali)-[~/Lab5]
└─$ python3 hmac_verify.py
Enter the received HMAC tag: 223e2b14c5fe1d878c436101936543d5b81bc55a20c94bc30d6cf2ba284fee59
Verification failed: Message has been altered or tag is invalid.
```

> Shows the third modification and verification failure.

Explanation

In every case, the verification script printed "Verification failed: Message has been altered or tag is invalid." This occurred because the HMAC tag is generated from the every single bit of the message. When the original message is altered, the new tag from the modified message no longer matches the original tag. We then identify that the message has been tampered with and we properly reject it

This experiment shops how MACs provide integrity protection; an attacker who intercepts our message and then modifies its contents cannot produce a proper valid tag for the altered message without knowing our secret key. Since our secret key is shared only by those engaging in legitimate communication, a validated tag gives the message receiver acknowledgement that the message was not altered and was written by someone possessing the shared secret key.

Task 6 – Generate HMAC Tags for Multiple Messages

What Was Done

A script `hmac_multi.py` was created to automatically compute HMAC tags for all three message files in a single run. This task demonstrates that the same key consistently produces different tags for different messages. One message was then modified slightly, and the script was rerun to confirm the tag change.

Script: hmac_multi.py

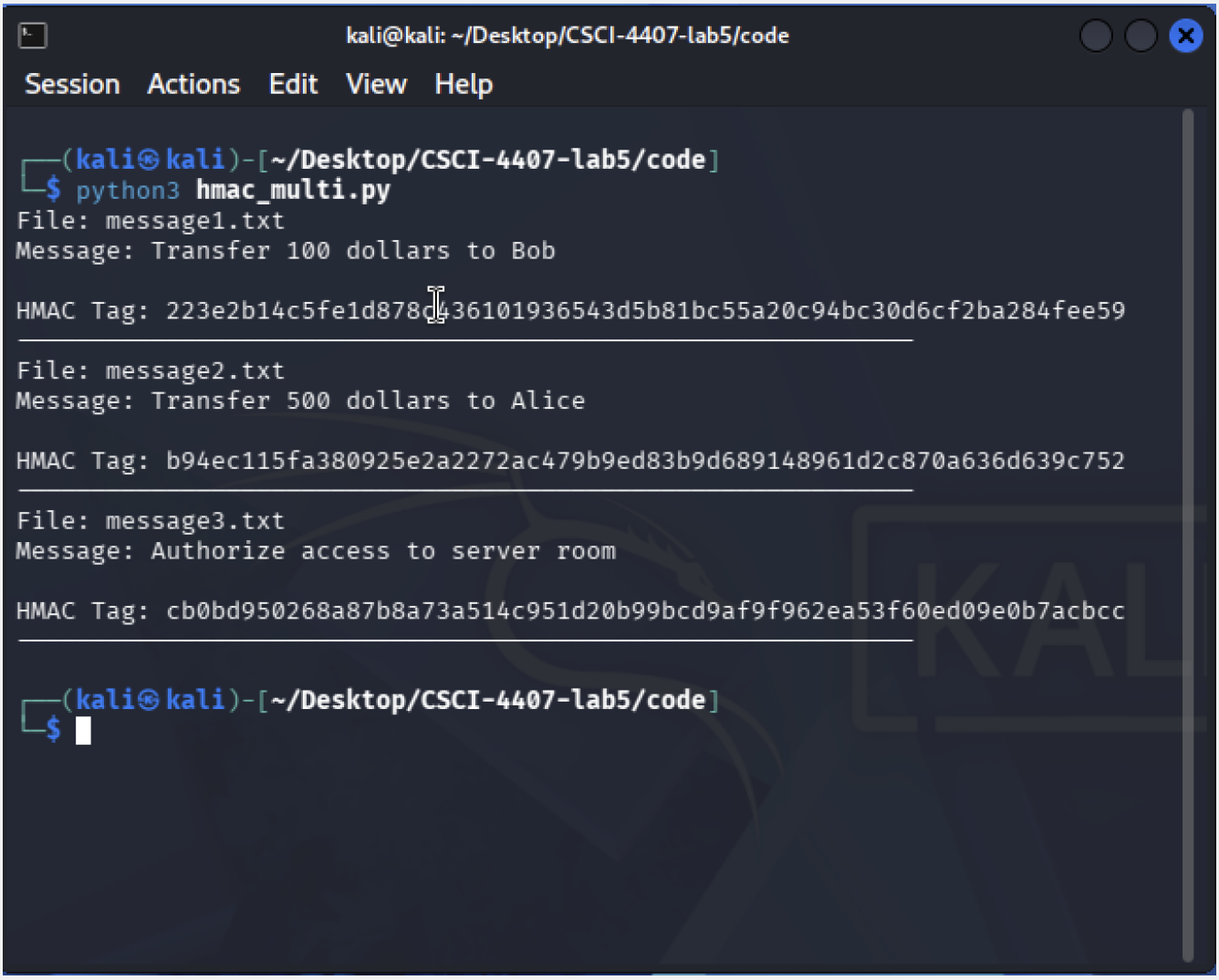
```python
import hmac
import hashlib

key = b"secretkey123"
files = ["message1.txt", "message2.txt", "message3.txt"]

for filename in files:
    with open(filename, "rb") as f:
        message = f.read()

    tag = hmac.new(key, message, hashlib.sha256).hexdigest()

    print("File:", filename)
    print("Message:", message.decode())
    print("HMAC Tag:", tag)
    print("-" * 60)
```

Commands Executed

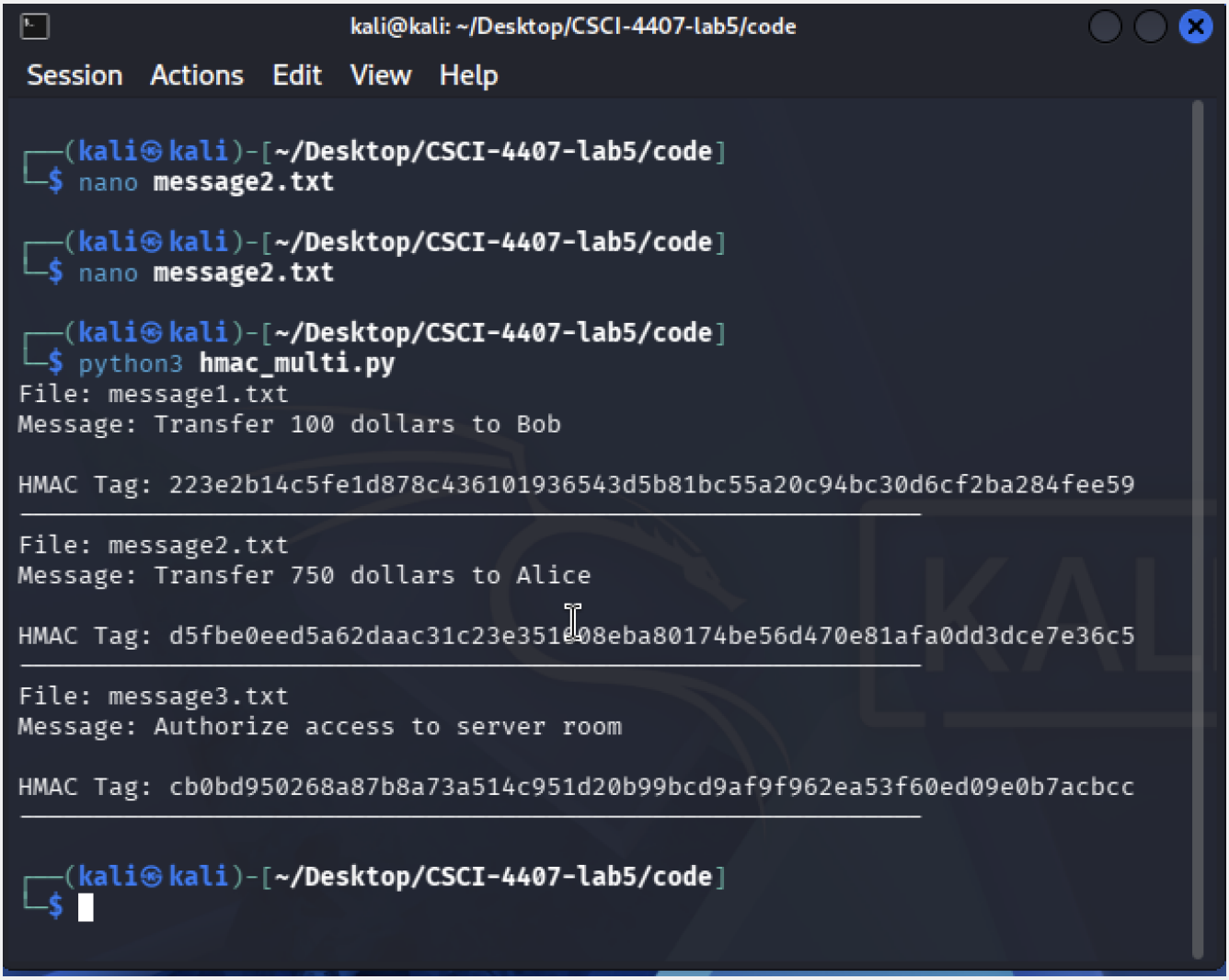
```
python3 hmac_multi.py

# Modify message2.txt slightly, then rerun
nano message2.txt
python3 hmac_multi.py
```

Screenshot Evidence

[SCREENSHOT: task6_multi_output.png] Shows execution of `hmac_multi.py` with HMAC tags for all three messages.



[SCREENSHOT: task6_modified_output.png] Shows execution after `message2.txt` was slightly modified, with the changed tag.



Comparison of Authentication Tags

Message File	Original Tag	Tag After Modification
message1.txt	223e2b14c5fe1d878c436101936543d5b81bc55a20c94bc30d6cf2ba284fee59	(unchanged)
message2.txt	b94ec115fa380925e2a272ac479b9ed83b9d689148961d2c870a636d639c752	d5fbe0eed5a62daac31c23e351e08eba80174be56d470e81afa0d
message3.txt	cb0bd950268a87b8a73a514c951d20b99bcd9af9f962ea53f60ed09e0b7acbcc	(unchanged)

Explanation

Each message produces a distinct HMAC tag even though the same key is used for all three. This is a consequence of HMAC's design: the message content is one of the inputs to the hash function, so different message inputs yield different hash outputs. This property is necessary for a MAC to be useful — if two different messages could produce the same tag, an attacker could substitute one message for another while keeping the valid tag.

After modifying a single word in `message2.txt`, the generated tag changed completely. This behavior prevents message forgery: an attacker who wants to tamper with a message and keep it valid would need to produce a new correct tag, which requires knowing the secret key. Even when the same key is used, different messages produce different HMAC tags because the message itself is part of the hash input, so any change in the message results in a completely different output.

Task 7 – Change the Secret Key and Observe the Effect

What Was Done

The secret key in `hmac_multi.py` was changed twice and the script was rerun each time using the same message files. This experiment demonstrates that the authentication tag depends on both the message and the key. Without the correct key, an attacker cannot generate valid tags or verify messages.

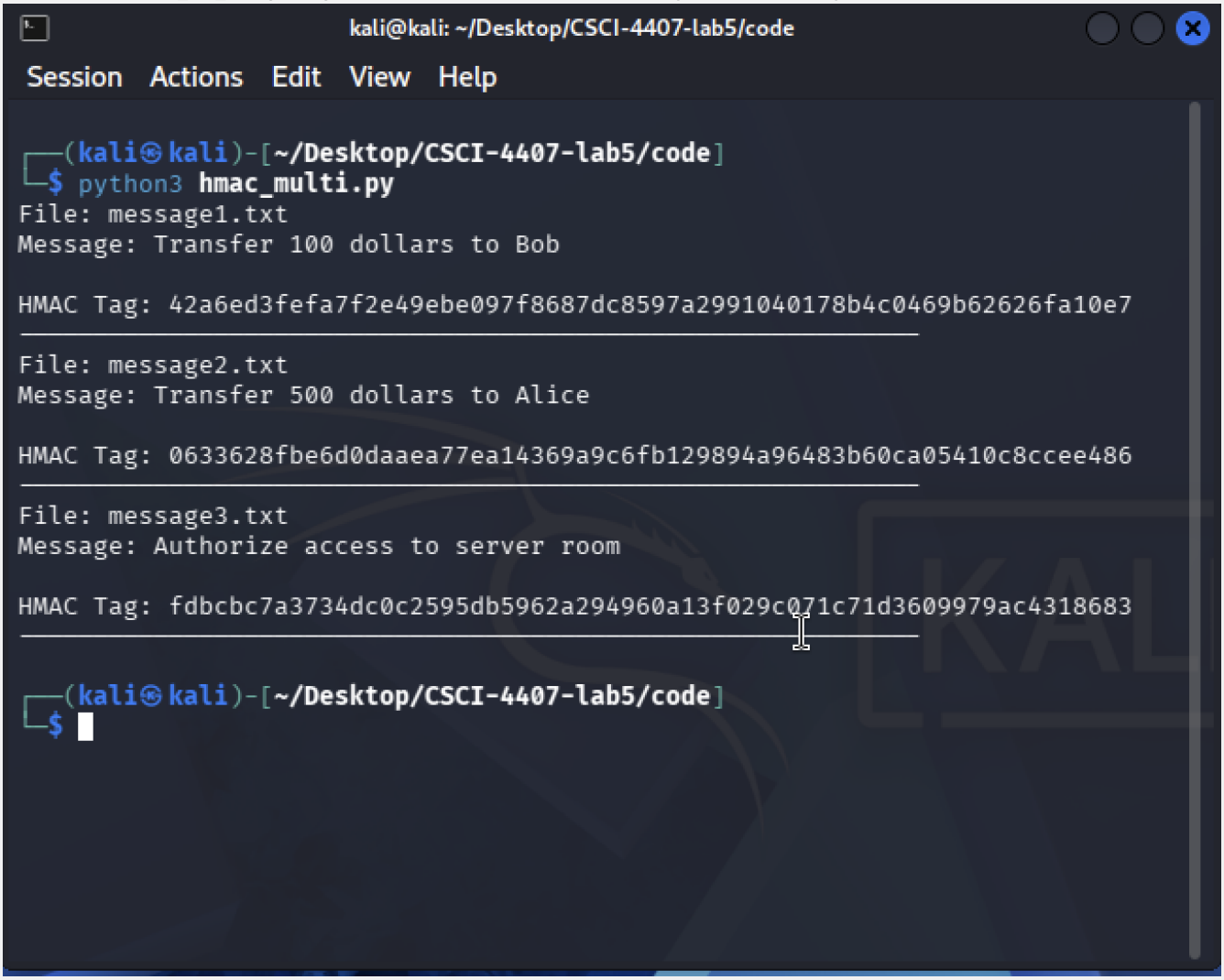
Commands Executed

```
# Key change 1: set key = b"newsecret456" in hmac_multi.py
nano hmac_multi.py
python3 hmac_multi.py
```

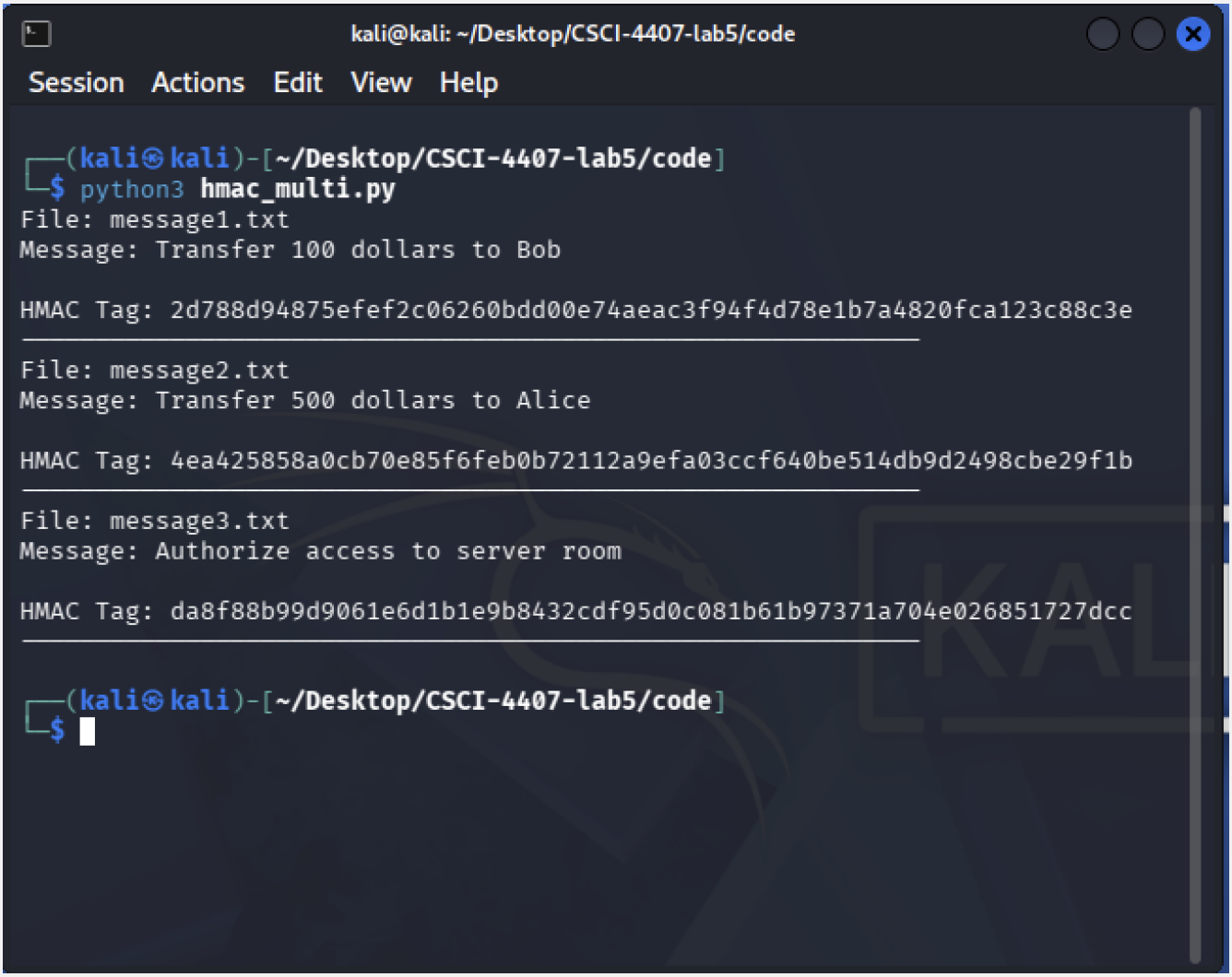
```
# Key change 2: set key = b"group10secret123" in hmac_multi.py
nano hmac_multi.py
python3 hmac_multi.py
```

Screenshot Evidence

[SCREENSHOT: task7_key_change1.png] Shows output with `newsecret456` — all tags differ from the originals.



[SCREENSHOT: task7_key_change2.png] Shows output with `group10secret123` — all tags differ again.



Comparison of Tags Across Keys

Comparison of Tags Across Keys

Message	Key: secretkey123	Key: newsecret456
message1.txt	223e2b14c5fe1d878c436101936543d5b81bc55a20c94bc30d6cf2ba284fee59	42a6ed3fefa7f2e49ebe097f8687dc8597a2991040178b4c0469b
message2.txt	b94ec115fa380925e2a272ac479b9ed83b9d689148961d2c870a636d639c752	0633628fbe6d0daaea77ea14369a9c6fb129894a96483b60ca05
message3.txt	cb0bd950268a87b8a73a514c951d20b99bcd9af9f962ea53f60ed09e0b7acbcc	fdbcbc7a3734dc0c2595db5962a294960a13f029c071c71d3609

Explanation

The HMAC construction incorporates the secret key at two levels: before and after the inner hash computation. As a result, the authentication tag is uniquely determined by the combination of key and message. When the key changes, the tag changes completely even if the message remains the same.

This property is critical for security. Because the tag depends on the key, an attacker who does not know the key cannot compute the correct tag for any message, including messages the attacker has never seen. Even if an adversary collects many valid `(M, T)` pairs, they cannot produce a valid tag for a new message without access to the secret key.

Keeping the secret key private is therefore the foundation of MAC security. If the key is compromised, the entire authentication mechanism fails, as any party with the key can generate valid tags for arbitrary messages. This proves that without the correct secret key, it is computationally infeasible to generate valid authentication tags, ensuring both message integrity and authenticity.

Task 8 – Replay Attack Experiment

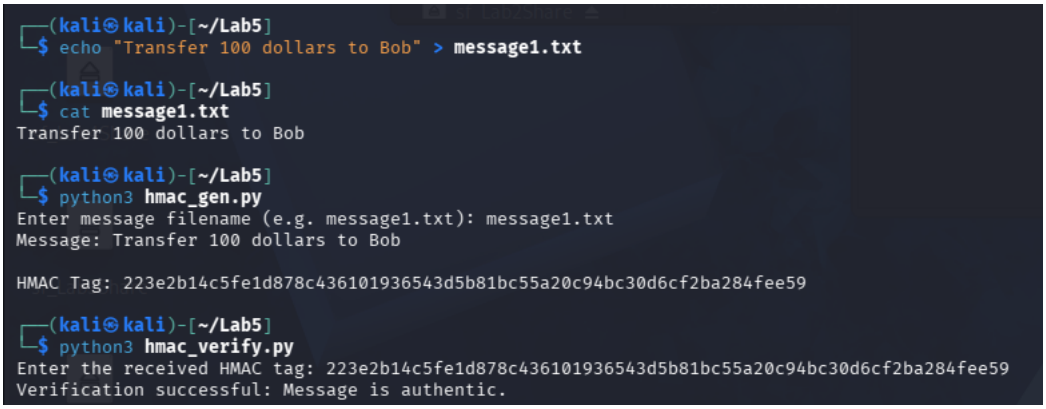
What Was Done

A replay attack was simulated to demonstrate a fundamental limitation of MACs: they verify message integrity and authenticity, but they do not guarantee **freshness**. An attacker who captures a valid `(M, T)` pair can resubmit it at any time and a MAC-only verifier will accept it, because the tag is still mathematically correct.

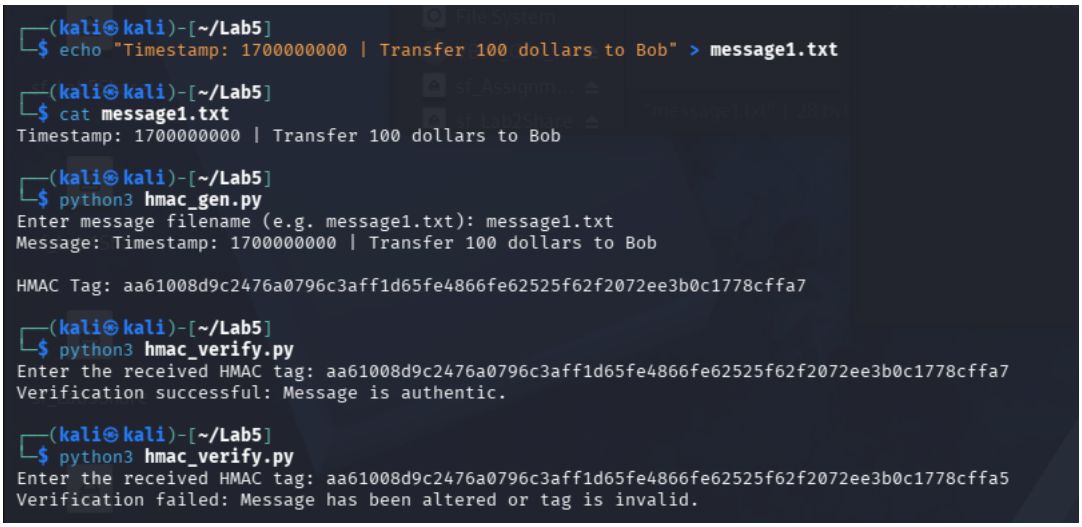
Commands Executed

```
# Restore original message
echo "Transfer 100 dollars to Bob" > message1.txt
cat message1.txt

# Generate tag
python3 hmac_gen.py

# First verification (legitimate)
python3 hmac_verify.py   # enter the tag above

# Second verification (replay)
python3 hmac_verify.py   # enter the same tag again

# Third verification (replay again)
python3 hmac_verify.py   # enter the same tag a third time

# Add timestamp to message
echo "Timestamp: 1700000000 | Transfer 100 dollars to Bob" > message1.txt
cat message1.txt
python3 hmac_gen.py

# Verify updated message
python3 hmac_verify.py   # enter the new tag
```

Screenshot Evidence



> Shows the restored message content. Shows the HMAC tag generated for the original message. **Recorded tag:** `223e2b14c5fe1d878c436101936543d5b81bc55a20c94bc30d6cf2ba284fee59` Shows "Verification successful" for the first (legitimate) verification.



> Shows the same message accepted again on the second submission — "Verification successful." Shows the message accepted a third time with the same tag — "Verification successful."

alt text

> Shows the updated message with embedded timestamp. Shows the new HMAC tag generated for the timestamped message. Shows successful verification of the timestamped message with its new tag.

Summary of Replay Results

Attempt	Message	Tag Used	Result

Attempt	Message	Tag Used	Result
First (legitimate)	Transfer 100 dollars to Bob	[223e2b14c5fe1d878c436101936543d5b81bc55a20c94bc30d6cf2ba284fee59]	Verification successful
Second (replay)	Transfer 100 dollars to Bob	[223e2b14c5fe1d878c436101936543d5b81bc55a20c94bc30d6cf2ba284fee59]	Verification successful
Third (replay)	Transfer 100 dollars to Bob	[223e2b14c5fe1d878c436101936543d5b81bc55a20c94bc30d6cf2ba284fee59]	Verification successful

Explanation: Why the Replay Succeeds

A MAC only checks whether the tag is correct for the contents of the message using the provided secret key. The verification process has no memory of previous transactions and therefore it does not store what message and key pairs have been processed before. Because the message and tag have not changed between replay attempts the MAC tag still matches the sent tag, so then verification succeeds every time. Due to the verifcation accepting the pairs over and over again the receiver accepts these messages every time without fail. MAC alone cannot gurantee freshness as it needs additional help to determine when the messages were sent and if they have been previously processed.

Discussion: Preventing Replay Attacks

To prevent replay attacks we need additional information that identifies which messages have been sent when, in what order, and if the message has been sent previously. This is done with Timestamps, Sequence numbers, and nonces; these all help MACs know if the message is being replayed or not. Timestamps allow the MAC to see the time the message was sent and check if it was sent recently or if it has been sent earlier and therefore is outdated then rejected by the system. Sequence numbers show what order the messages have been sent by incrementing the message counter, the reciever tracks this counter and rejects any message that is not higher than the last message sent on the counter. Nonces are a randomly generated value that the reciever generates and can be used to challenge messages that are sent to the reciever. This Nonce is used by the reciever to see if the sender is authentic by seeing if the nonce is included in the next message, as all nonces are generated per session an old nonce will not be the same as the new nonce and a replayed message containing an old unsed nonce will then have their message rejected.

Task 9 – Implementing a PRF-Based Authentication Function

What Was Done

A custom PRF-based authentication script `prf_experiment.py` was implemented from scratch using `hashlib.sha256` directly rather than Python's `hmac` library. The function follows the construction `F_K(M) = SHA-256(K || M)`, where the secret key and message are concatenated before hashing. Three experiments were run: generating tags for three different messages under the same key, modifying one message and observing the tag change, and changing the key entirely and observing that all tags change.

Script: prf_experiment.py

```python
import hashlib

def prf_authenticate(key: bytes, message: bytes) -> str:
    """
    Compute a PRF-based authentication tag for a message.
    F_K(M) = SHA-256(K || M)
    """
    combined = key + message
    digest = hashlib.sha256(combined).hexdigest()
    return digest


def run_experiment(secret_key: bytes, messages: list):
    print(f"Secret Key: {secret_key.decode()}")
    print("=" * 60)
    for msg in messages:
        tag = prf_authenticate(secret_key, msg.encode())
        print(f"Message : {msg}")
        print(f"Auth Tag: {tag}")
        print("-" * 60)
    print()


# Experiment 1: Three messages with the same key
print("=== Experiment 1: Three Messages, Same Key ===")
key1 = b"myGroupKey789"
messages = [
    "Transfer 100 dollars to Bob",
    "Transfer 500 dollars to Alice",
    "Authorize access to server room",
]
run_experiment(key1, messages)

# Experiment 2: Modify one message
print("=== Experiment 2: Modified Message ===")
modified_messages = [
    "Transfer 100 dollars to Bob",
    "Transfer 999 dollars to Bob",
```

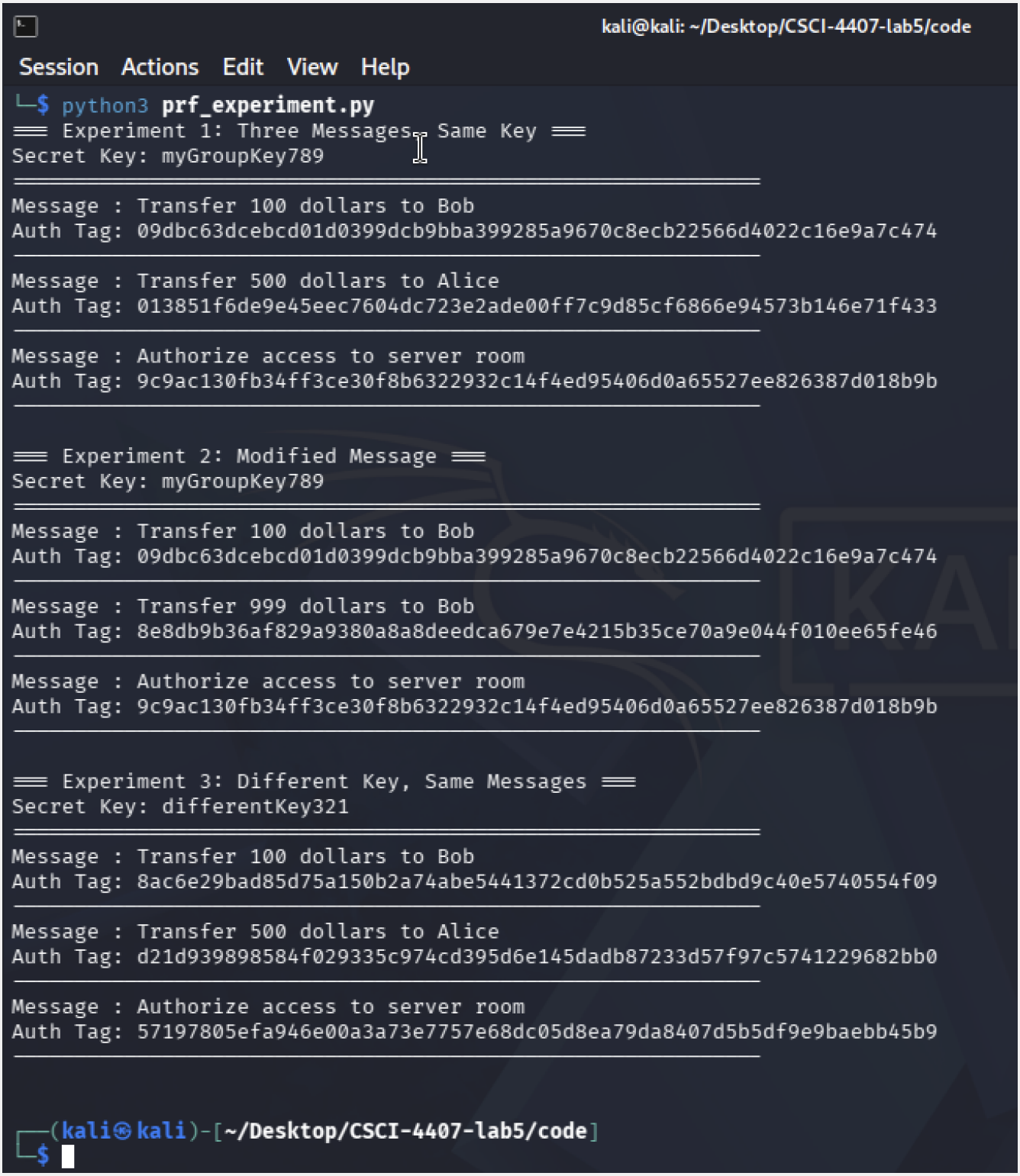
```
    "Authorize access to server room",
]
run_experiment(key1, modified_messages)

# Experiment 3: Different key, same messages
print("=== Experiment 3: Different Key, Same Messages ===")
key2 = b"differentKey321"
run_experiment(key2, messages)
```

Commands Executed

```
python3 prf_experiment.py
```

Screenshot Evidence

[SCREENSHOT: task9.png] Shows the full `prf_experiment.py` script in the terminal.



[SCREENSHOT: task9experiment1.png] Shows authentication tags for all three original messages using `myGroupKey789`.

```
═══ Experiment 1: Three Messages, Same Key ═══
Secret Key: myGroupKey789
═══════════════════════════════════════════════════════════
Message : Transfer 100 dollars to Bob
Auth Tag: 09dbc63dcebcd01d0399dcb9bba399285a9670c8ecb22566d4022c16e9a7c474
───────────────────────────────────────────────────────────
Message : Transfer 500 dollars to Alice
Auth Tag: 013851f6de9e45eec7604dc723e2ade00ff7c9d85cf6866e94573b146e71f433
───────────────────────────────────────────────────────────
Message : Authorize access to server room
Auth Tag: 9c9ac130fb34ff3ce30f8b6322932c14f4ed95406d0a65527ee826387d018b9b
───────────────────────────────────────────────────────────
```

[SCREENSHOT: task9experiment2.png] Shows how the tag for the second message changes after modifying the amount from 500 to 999.

```
═══ Experiment 2: Modified Message ═══
Secret Key: myGroupKey789
═══════════════════════════════════════════════════════════
Message : Transfer 100 dollars to Bob
Auth Tag: 09dbc63dcebcd01d0399dcb9bba399285a9670c8ecb22566d4022c16e9a7c474
───────────────────────────────────────────────────────────
Message : Transfer 999 dollars to Bob
Auth Tag: 8e8db9b36af829a9380a8a8deedca679e7e4215b35ce70a9e044f010ee65fe46
───────────────────────────────────────────────────────────
Message : Authorize access to server room
Auth Tag: 9c9ac130fb34ff3ce30f8b6322932c14f4ed95406d0a65527ee826387d018b9b
───────────────────────────────────────────────────────────
```

[SCREENSHOT: task9experiment3.png] Shows how all tags change when the key is switched to `differentKey321`.

```
═══ Experiment 3: Different Key, Same Messages ═══
Secret Key: differentKey321
═══════════════════════════════════════════════════════════
Message : Transfer 100 dollars to Bob
Auth Tag: 8ac6e29bad85d75a150b2a74abe5441372cd0b525a552bdbd9c40e5740554f09
───────────────────────────────────────────────────────────
Message : Transfer 500 dollars to Alice
Auth Tag: d21d939898584f029335c974cd395d6e145dadb87233d57f97c5741229682bb0
───────────────────────────────────────────────────────────
Message : Authorize access to server room
Auth Tag: 57197805efa946e00a3a73e7757e68dc05d8ea79da8407d5b5df9e9baebb45b9
───────────────────────────────────────────────────────────
```

Recorded Results

Experiment 1 – Original Messages, Key: myGroupKey789

Message	Auth Tag
Transfer 100 dollars to Bob	09dbc63dcebdc01d0399dcb9bba399285a9670c8ecb22566d4022c16e9a7c474
Transfer 500 dollars to Alice	013851f6de9e45eec7604dc723e2ade00ff7c9d85cf6866e94573b146e71f433
Authorize access to server room	9c9ac130fb34ff3ce30f8b6322932c14f4ed95406d0a65527ee826387d018b9b

Experiment 2 – Modified Message 2, Same Key

Message	Auth Tag
Transfer 100 dollars to Bob	09dbc63dcebdc01d0399dcb9bba399285a9670c8ecb22566d4022c16e9a7c474
Transfer 999 dollars to Bob	8e8db9b36af829a9380a8a8deedca679e7e4215b35ce70a9e044f010ee65fe46
Authorize access to server room	9c9ac130fb34ff3ce30f8b6322932c14f4ed95406d0a65527ee826387d018b9b

Experiment 3 – Original Messages, Key: differentKey321

Message	Auth Tag
Transfer 100 dollars to Bob	8ac6e29bad85d75a150b2a74abe5441372cd0b525a552bdbd9c40e5740554f09
Transfer 500 dollars to Alice	d21d939898584f029335c974cd395d6e145dadb87233d57f97c5741229682bb0
Authorize access to server room	57197805efa946e00a3a73e7757e68dc05d8ea79da8407d5b5df9e9baebb45b9

Analysis Questions

What happens to the authentication output when the message changes? When the message changes, even by a single character, the SHA-256 hash of the concatenated input `(K || M)` changes completely. This is the avalanche effect: a small difference in the input produces an entirely different output, making it impossible to predict the new tag from the old one.

What happens when the secret key changes? Changing the key changes the input to the hash function from `K || M` to `K' || M`. Because `K ≠ K'`, the concatenated input is different, and the SHA-256 output changes completely for every message. The entire set of previously valid tags becomes invalid when the key changes.

Why would it be difficult for an attacker to compute the correct output without knowing the key? SHA-256 is a one-way function: given the output, it is computationally infeasible to recover the input. An attacker who does not know `K` cannot compute `SHA-256(K || M)` because they do not have access to `K`. Even if the attacker observes many valid `(M, F_K(M))` pairs, they cannot work backwards to determine `K` or predict the correct tag for any new message.

Why can a PRF be used to build a message authentication mechanism? A pseudorandom function produces outputs that are unpredictable to anyone who does not know the secret key. This property is useful for authentication: if the output of `F_K(M)` cannot be predicted without knowing `K`, then an attacker cannot forge a valid tag for a new message. Our experiment demonstrated this behavior — changing either the message or the key produced a completely different output, and there is no obvious pattern that would allow an attacker to compute the next tag without the key.

Relationship Between PRFs and MACs

While `F_K(M) = SHA-256(K || M)` is a simple keyed hash construction and not the standardized HMAC design, it still demonstrates the key idea behind PRF-style authentication: the output depends on both the secret key and the message. In our experiment, changing either the message or the key produced a completely different output, which illustrates why keyed functions can support message authentication.

More generally, secure PRFs are closely related to MACs because a keyed function with unpredictable outputs can be used to authenticate messages. This lab exercise demonstrates that relationship conceptually, even though practical systems typically use standardized constructions such as HMAC rather than raw key-message concatenation.

Conclusion

This lab provided hands-on experience with message authentication codes and pseudorandom functions. Through the nine tasks, the following core concepts were demonstrated:

- **Integrity and authenticity:** HMAC generates a tag that binds a message to a secret key. Any modification to the message invalidates the tag, allowing the receiver to detect tampering.
- **Unforgeability:** Without the secret key, an attacker cannot generate a valid tag for any message, including forged or modified messages.
- **Key dependence:** Changing the secret key completely changes all authentication tags, demonstrating that the security of the scheme rests entirely on the secrecy of the key.
- **Freshness limitation:** A MAC alone does not protect against replay attacks. Including a timestamp, counter, or nonce inside the authenticated message is necessary to guarantee message freshness.
- **PRF–MAC relationship:** A secure PRF can serve as a MAC. The custom `prf_experiment.py` script demonstrated this relationship using the construction `F_K(M) = SHA-256(K || M)`, showing that outputs are key-dependent and message-dependent even in a simple hand-built implementation.